

Total No. of Questions : 8]

SEAT No. :

PD-4502

[Total No. of Pages : 2

[6403]-311

**T.E. (Cyber Security Honour Course)
INFORMATION AND CYBER SECURITY
(2019 Pattern) (Semester - V) (310401)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 Or Q.2, Q.3, or Q.4, Q.5 or Q.6, Q.7, or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data if necessary.
- 5) Use of Calculator is allowed.

Q1) a) Explain SHA-1 in detail and explain how can mail security be achieved? [9]

b) Explain RSA algorithm. Perform encryption and decryption using RSA algorithm for following values $p=17$, $q=11$, $e=7$, $M=2$. [9]

OR

Q2) a) Explain Diffie-Hellman with justification. Give the details of attack on Diffie Hellman algorithm. [9]

b) Explain Elliptic curve Arithmetic and Elliptic Curve cryptography in details. [9]

Q3) a) Illustrate with a neat diagram components of risk identification and risk assessment. [8]

b) Explain in detail Indian legal perspective on cybercrimes. [9]

OR

Q4) a) What are the types of Laws? Differentiate between Law and Ethics in information security. [8]

b) List and explain any four commandments of computer ethics. [9]

Q5) a) Explain Kerberos. What four requirements were defined for Kerberos? [9]

b) What is the role of X.509 in cryptography? [9]

OR

P.T.O.

- Q6) a)** Discuss SSL with respect to four phases [9]
- Establish security capabilities
 - Server authentication and key exchange
 - Client authentication and key exchange
 - Finish

- b) Explain IP security protocols AH and IPS in detail. [9]

- Q7) a)** Define list and explain different types of worms. [8]

- b) What is an Intrusion Detection System? Describe types of IDS based on method of Detection? [9]

OR

- Q8) a)** What is a phishing attack explain with an example. What are the different types of phishing? [8]

- b) Explain following terms - (any 3) [9]

- i) SQL injection
- ii) Spyware
- iii) Adware
- iv) Ransomware

